

## Contribution Validation Report — ONE simulation run only

**Case:** all contributions matched

**Type:** Matched validation — all deterministic effects matched (expect mismatch  $< 0.05$  m)

Duration: 600 s

Towers: 5

Receivers: 1

Max PR meas/epoch: 5

Max EKF rows/epoch: 5

Final pos error: 5.7886 m

Pos RMS (last 20%): 5.0622 m

Mean NIS: 5.02 (E[NIS]  $\sim 5$ )

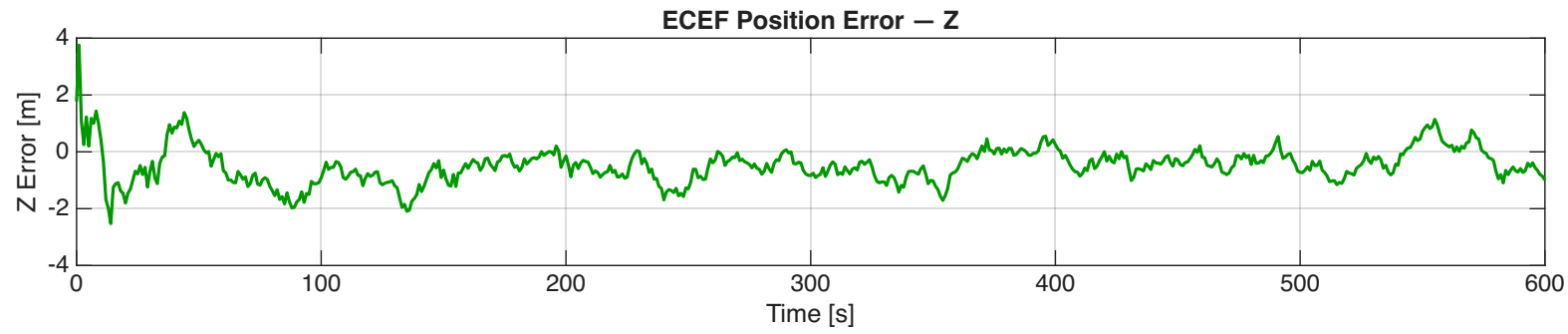
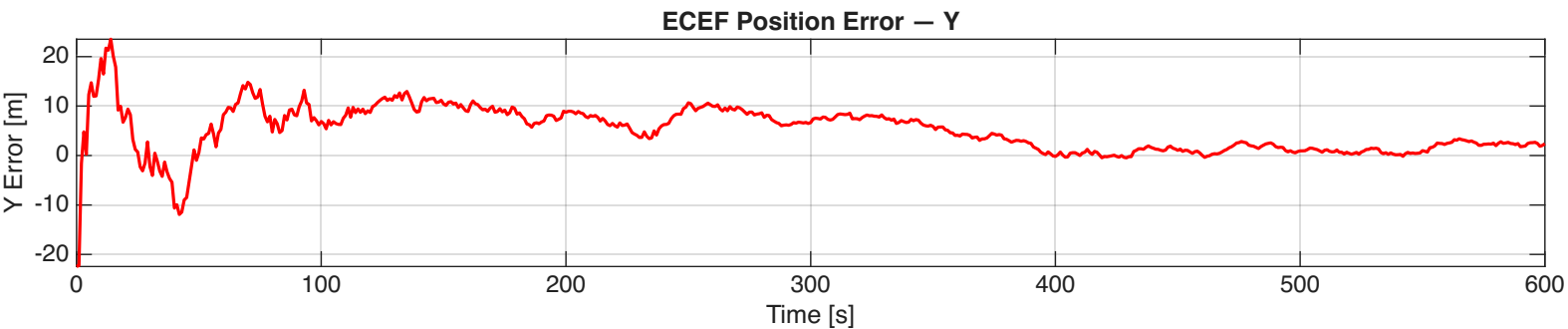
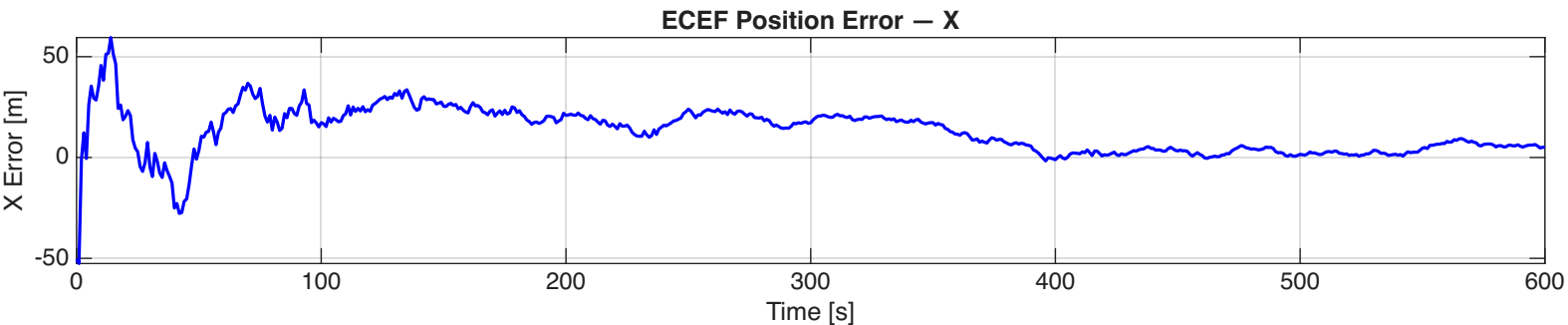
**Truth effects:** Sagnac, Shapiro, Trop, Iono, TowerSurvey, PCO, PCV

**Model effects:** Sagnac, Shapiro, Trop, Iono, TowerSurvey, PCO, PCV

**Mismatched:** none

**Output PDF:** /Users/ludwigmatuschka/Library/CloudStorage/OneDrive-Persönlich/Dokumente/ICH/Karriere/Cranfield/IRP/Codes/IRP\_simulation/oo\_v1/output/effect\_contribution\_validation\_report.pdf

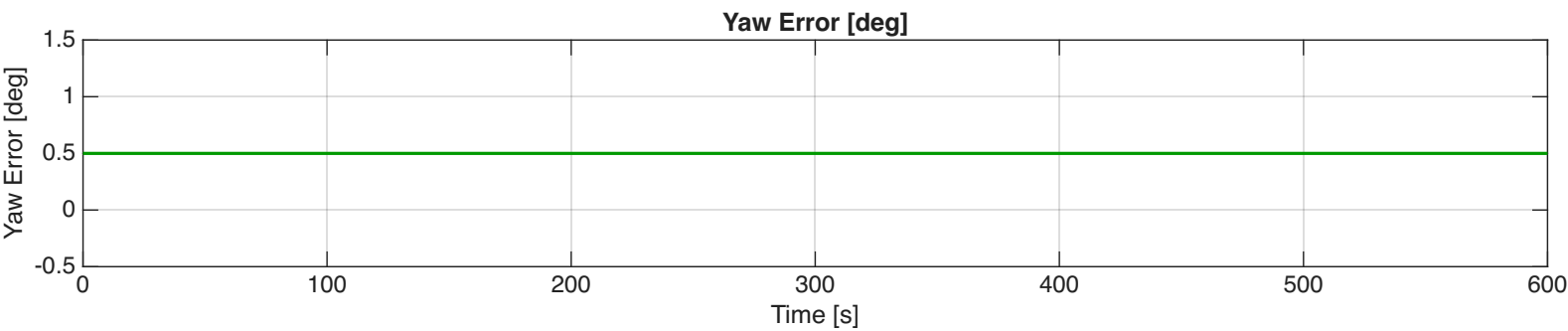
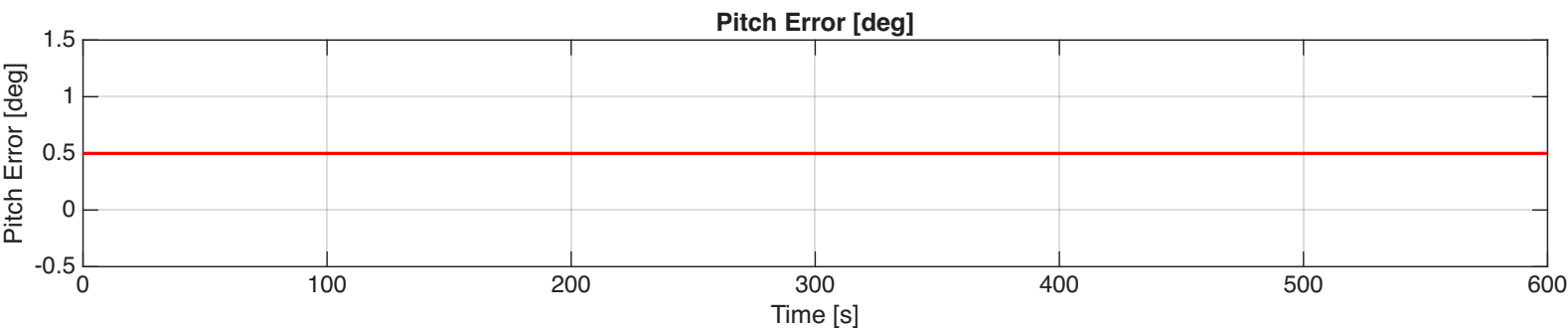
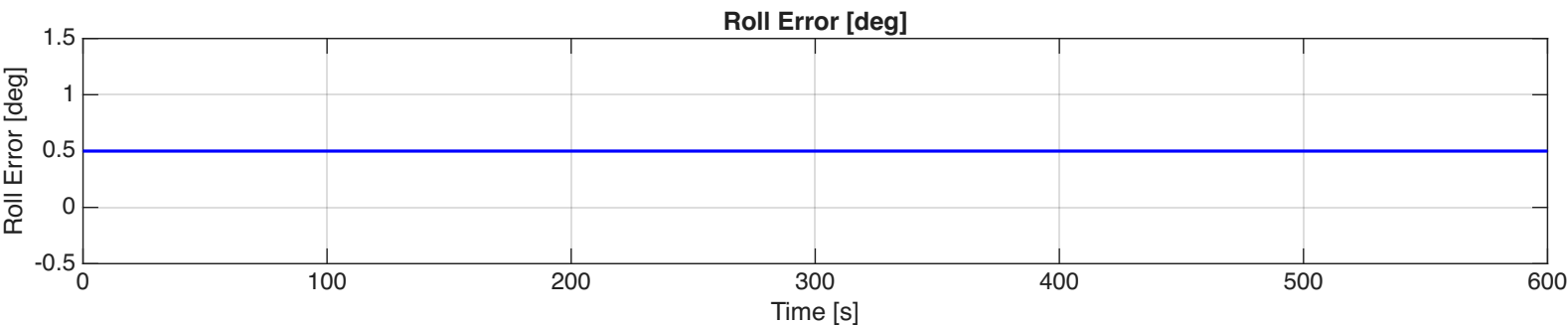
# Position Error Components (ECEF)



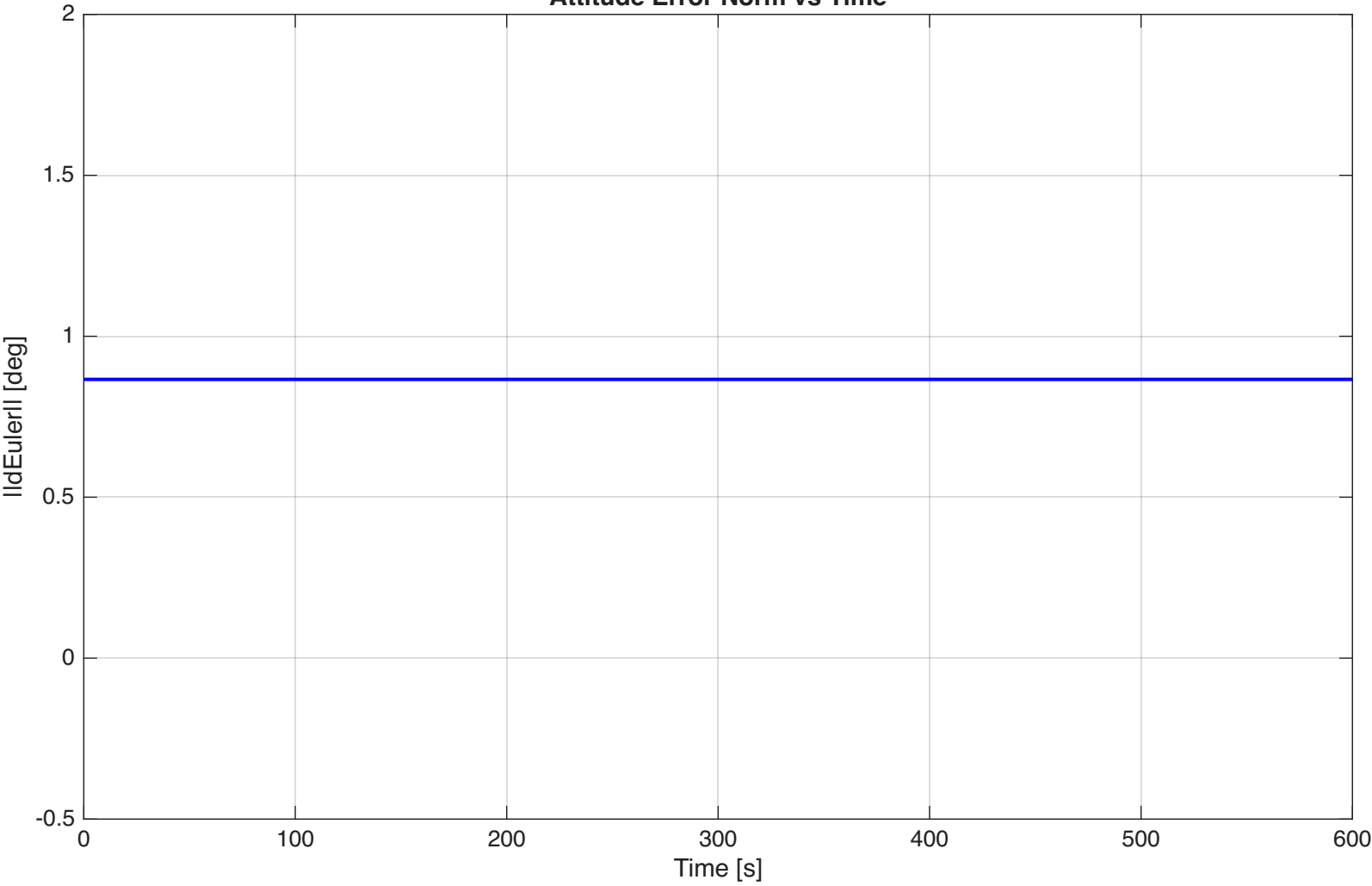
Position Error Norm vs Time



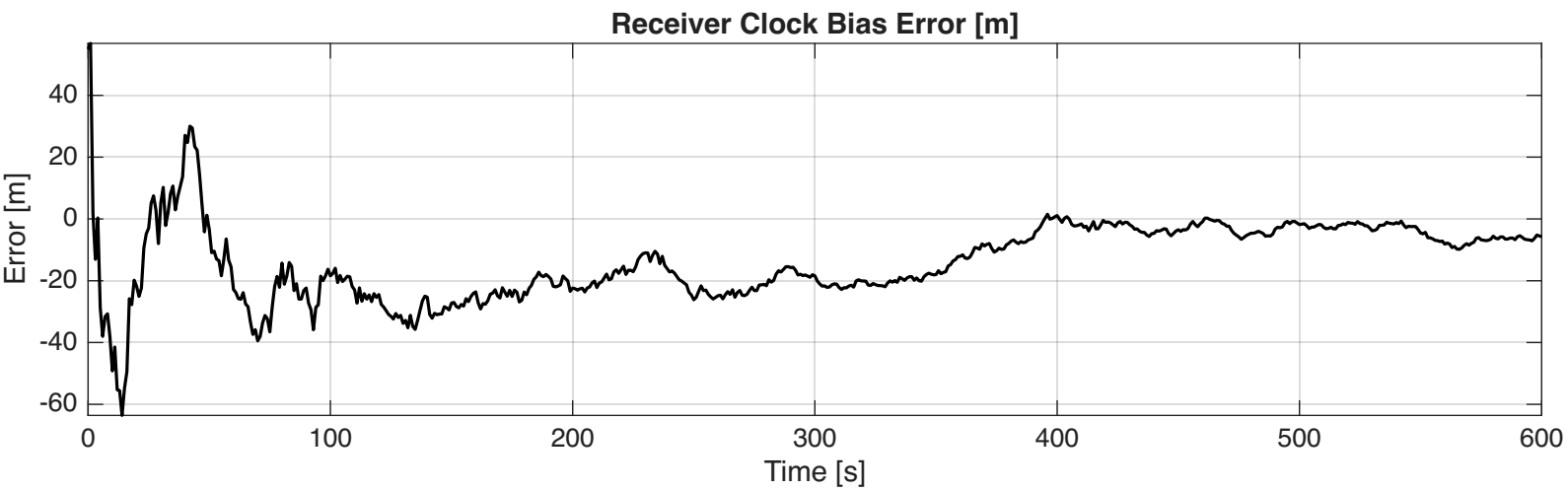
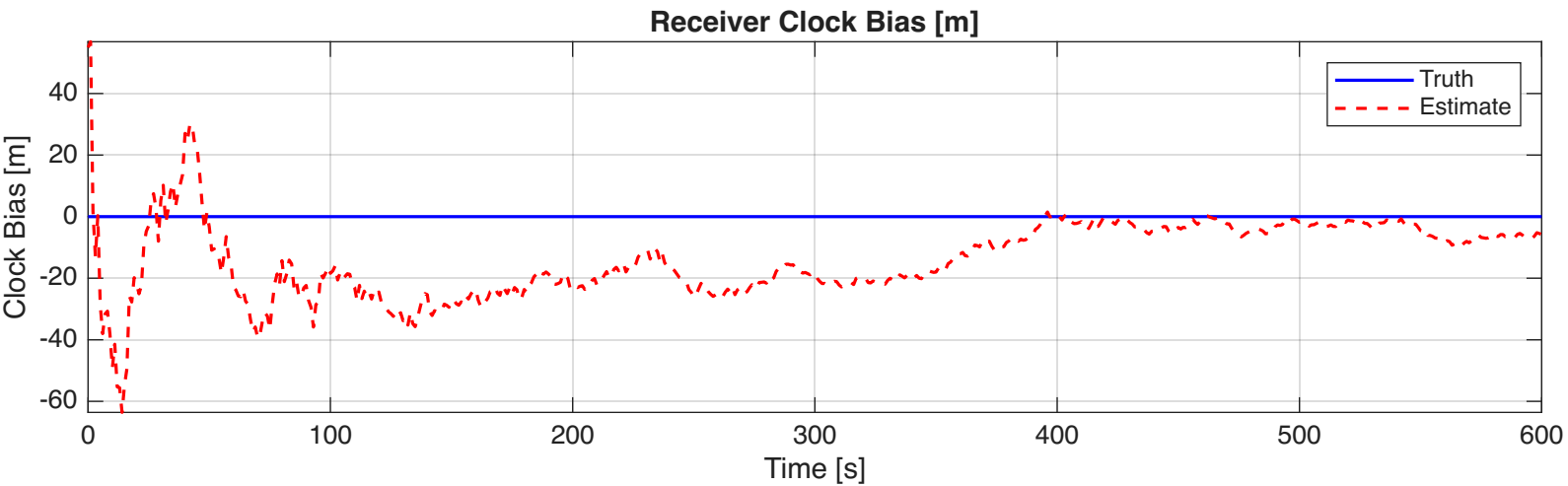
# Attitude Error Components



Attitude Error Norm vs Time

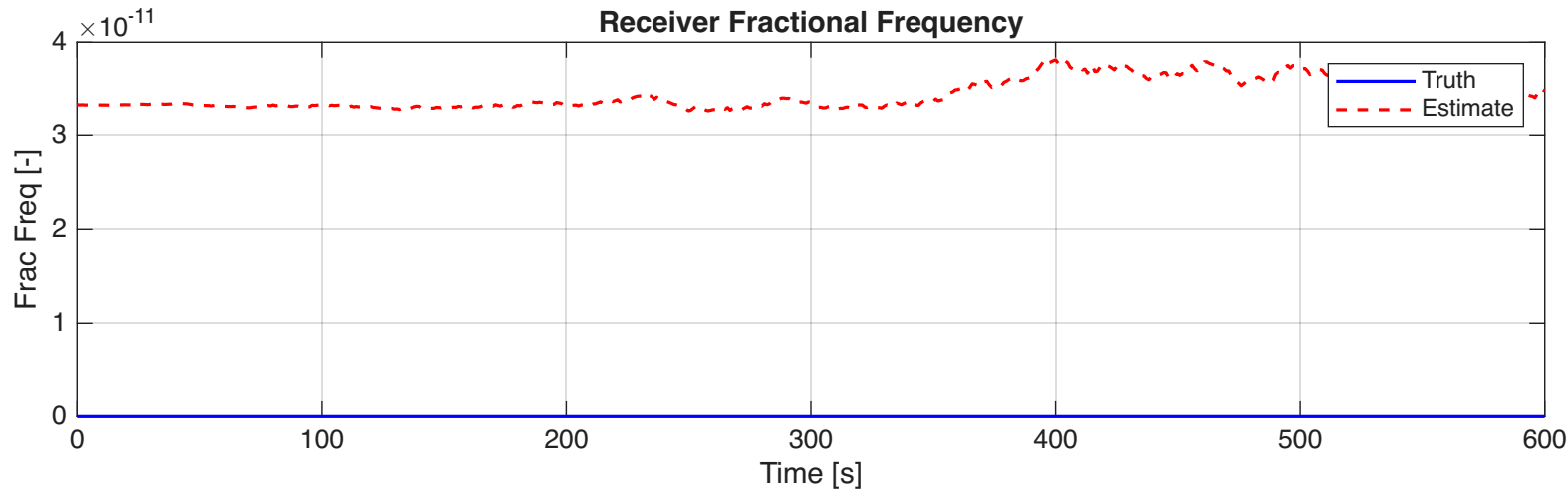


# Receiver Clock Bias

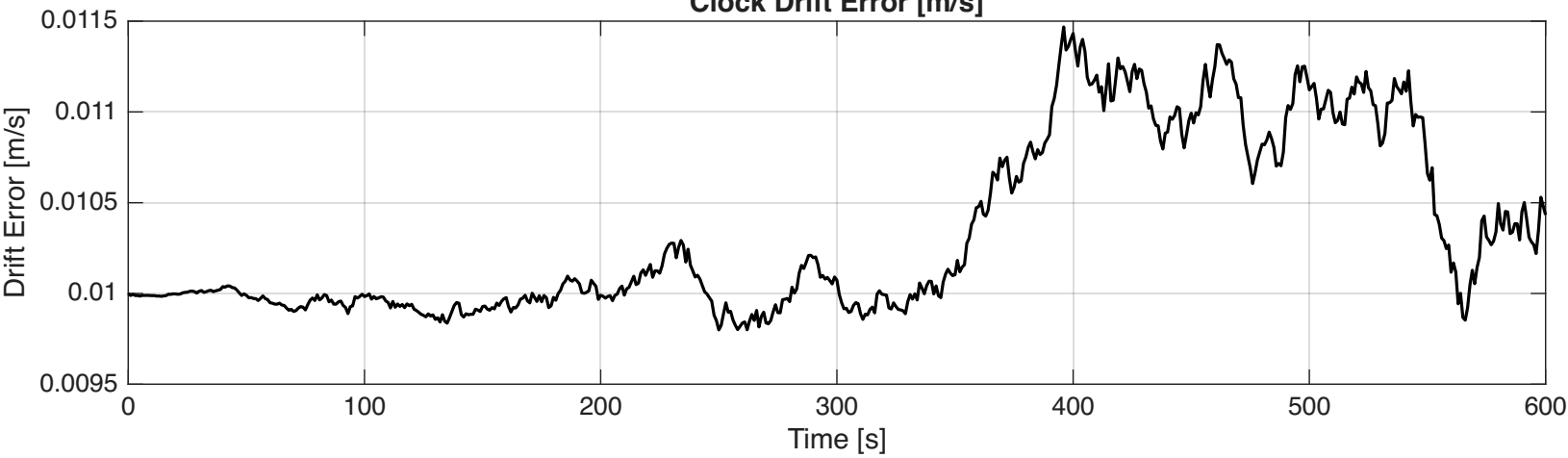


# Receiver Clock Drift / Fractional Frequency

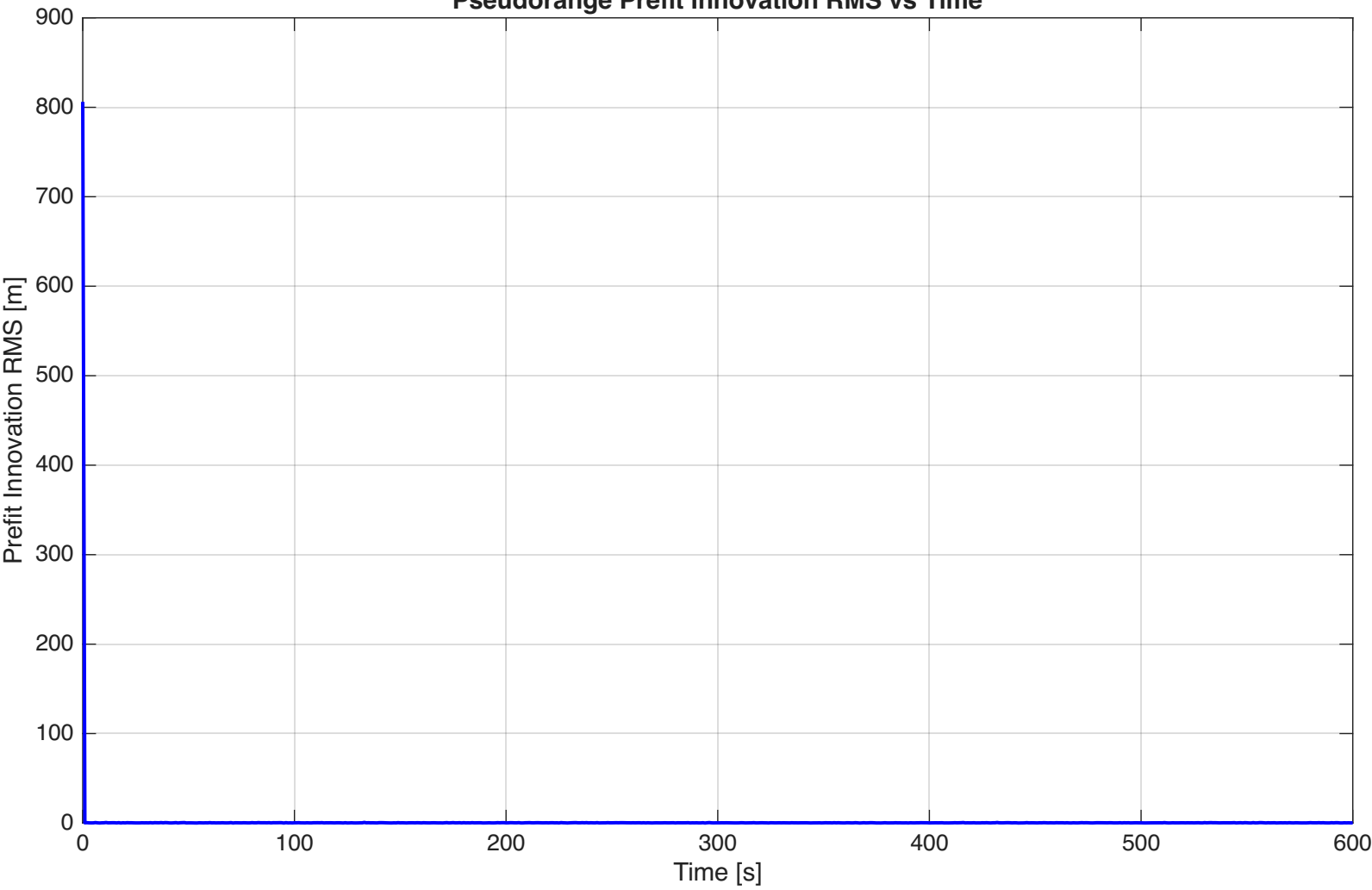
## Receiver Fractional Frequency



## Clock Drift Error [m/s]

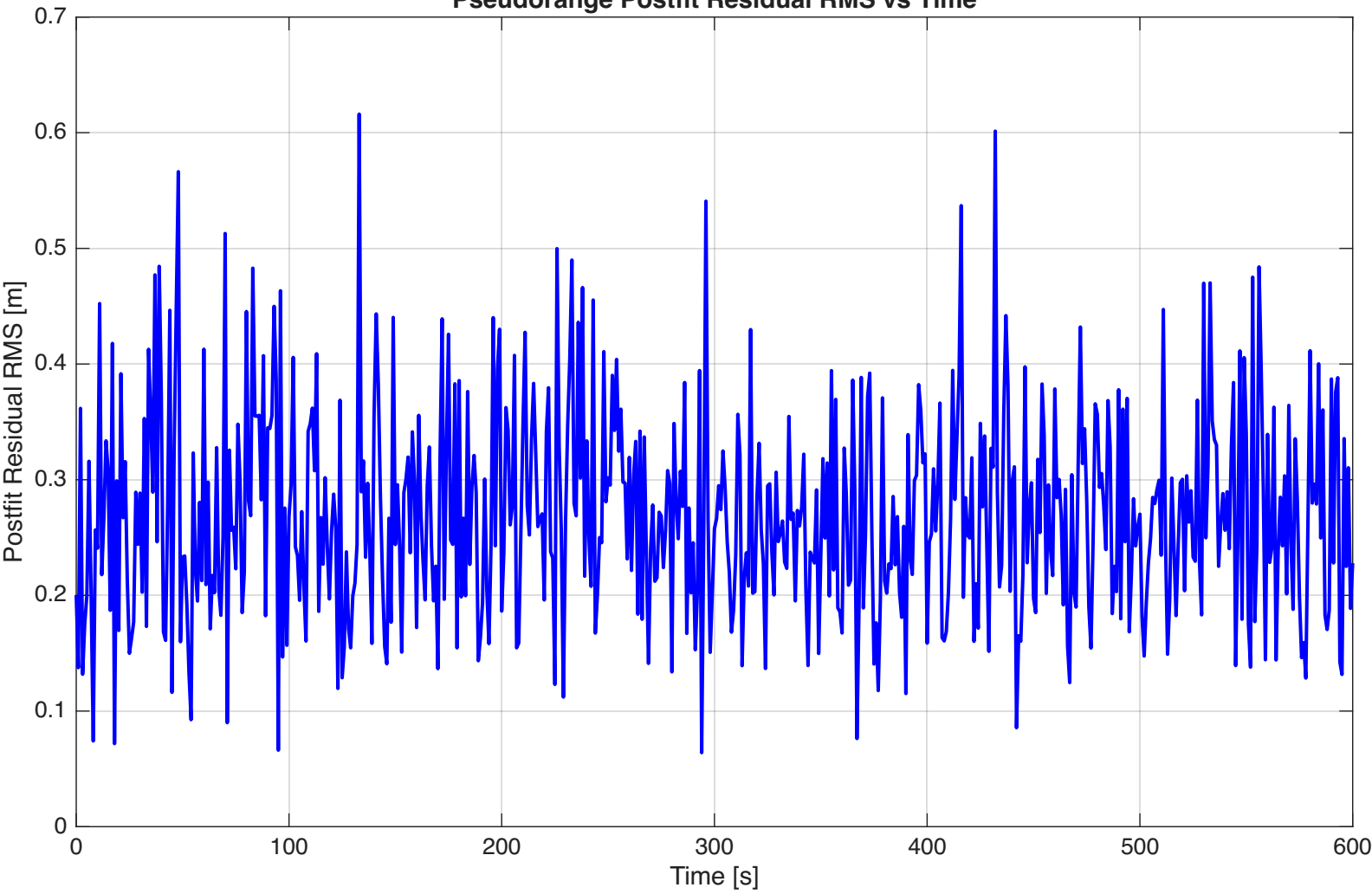


Pseudorange Prefit Innovation RMS vs Time

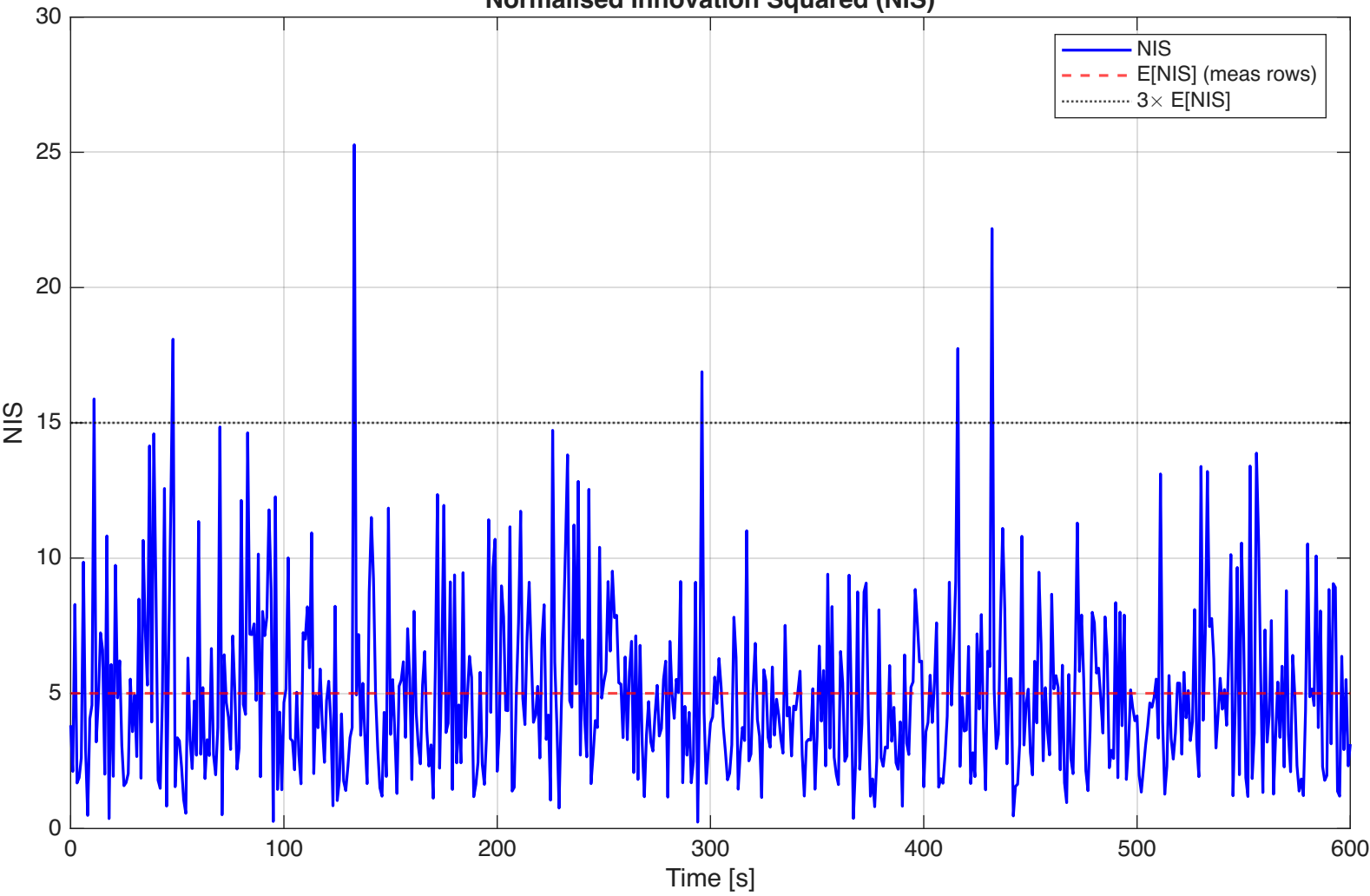




Pseudorange Postfit Residual RMS vs Time

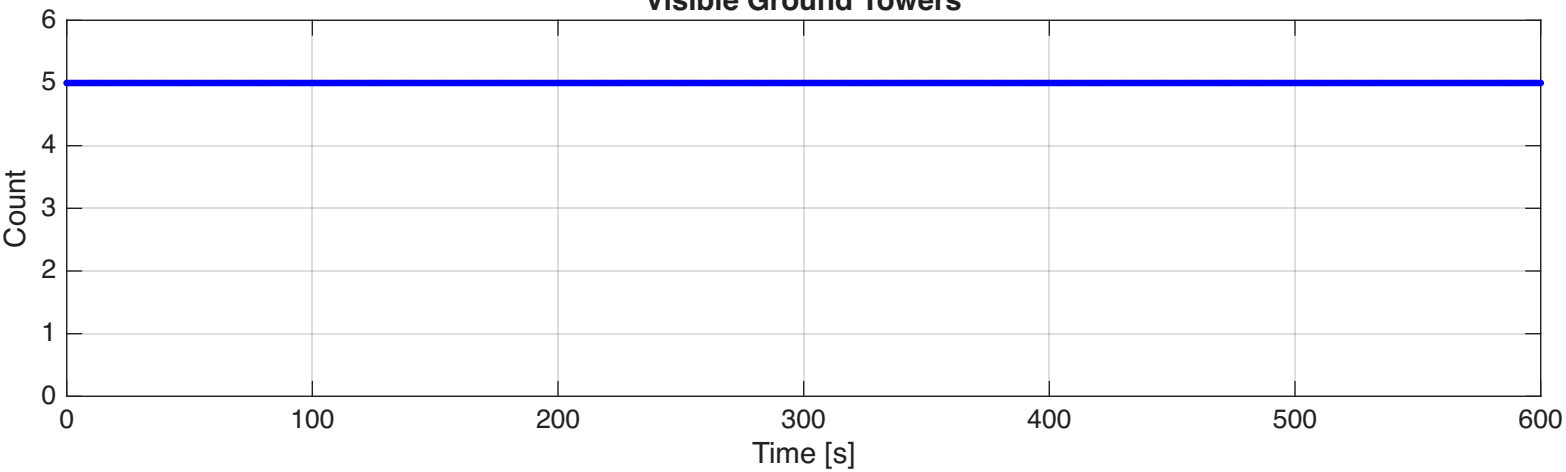


Normalised Innovation Squared (NIS)

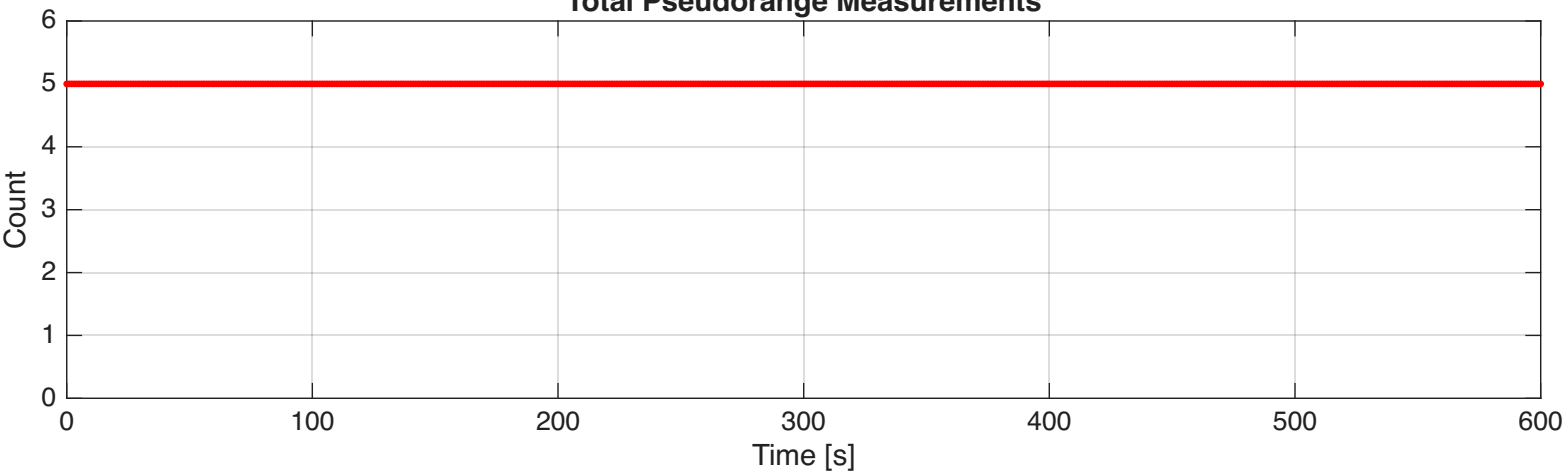


# Observations per Epoch

## Visible Ground Towers



## Total Pseudorange Measurements



Per-Source Error Contribution RMS (Truth - Model)

